

WINTON, Australia

WMO: 943420

Lat: 22.3617S

Lon: 143.0836E

Elev: 193

StdP: 99.03

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	7.4	-10.9	1.5	19.3	-8.9	1.8	20.0	9.8	19.3	9.2	18.6	2.4	90	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	13.1	41.0	20.7	39.8	20.7	38.7	20.7	25.4	31.6	24.9	31.1	24.4	30.9	4.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	19.6	27.0	23.6	18.8	26.9	22.8	18.0	27.0	79.4	31.3	77.1	31.1	74.9	30.7	32.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.9	8.0	DB	3.1	43.2	1.0	1.6	2.4	44.4	1.7	45.3	1.2	46.2	0.4	47.3	(4)
(5)			WB	-0.3	27.2	1.3	1.6	-1.2	28.4	-2.0	29.3	-2.7	30.2	-3.6	31.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	24.9	30.7	29.9	28.5	25.1	20.8	17.5	17.3	18.8	23.3	27.0	29.2	30.7	(6)	
	DBStd	5.77	2.67	2.60	2.35	2.53	3.01	3.24	3.02	3.08	3.27	3.15	2.78	2.82	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	159	0	0	0	1	12	54	57	32	3	1	0	0	(9)	
	CDD10.0	5425	642	557	573	454	336	226	225	271	398	526	576	643	(10)	
	CDD18.3	2542	383	324	314	204	89	30	24	45	151	268	326	384	(11)	
	CDH23.3	33771	5498	4456	4009	2228	841	269	286	636	1902	3553	4513	5579	(12)	
(13)	CDH26.7	19049	3372	2634	2218	1062	253	45	49	196	929	2038	2744	3508	(13)	
(14)	Wind	WSAvg	3.7	3.9	3.8	3.7	3.5	3.5	3.5	3.5	3.7	4.0	4.0	4.0	(14)	
(15) Precipitation	PrecAvg	396	73	93	48	21	20	13	13	7	7	18	34	58	(15)	
	PrecMax	1171	563	464	245	181	98	95	113	49	35	122	243	197	(16)	
	PrecMin	53	0	4	0	0	0	0	0	0	0	0	0	1	(17)	
	PrecStd	192	93	100	58	37	28	22	26	11	10	26	53	45	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.4	41.6	39.5	36.1	33.0	29.9	30.1	33.4	37.6	40.4	41.4	42.6	(19)	
		MCWB	21.4	21.3	21.1	20.1	18.6	18.0	17.5	17.3	18.4	18.9	20.1	20.9	(20)	
	2%	DB	40.5	39.5	37.9	34.6	30.8	28.0	27.8	30.8	35.4	38.4	39.8	41.0	(21)	
		MCWB	21.5	22.0	21.2	19.6	18.0	17.0	16.0	16.3	17.6	19.0	19.8	20.5	(22)	
	5%	DB	38.9	37.9	36.4	33.3	29.4	26.5	26.4	28.8	33.7	36.8	38.4	39.7	(23)	
		MCWB	21.7	22.0	21.1	19.2	17.4	16.4	15.3	15.2	17.0	18.4	19.5	20.5	(24)	
	10%	DB	37.5	36.5	35.0	32.1	27.9	24.8	24.8	27.0	31.9	35.2	37.0	38.2	(25)	
		MCWB	21.8	22.0	20.9	18.8	17.0	15.3	14.4	14.5	16.4	18.1	19.4	20.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.3	26.0	25.6	23.9	21.3	19.7	20.4	19.5	21.8	22.8	24.5	25.8	(27)	
		MCDB	32.7	31.7	29.8	30.0	27.5	26.8	23.5	28.3	28.4	31.4	32.3	34.0	(28)	
	2%	WB	25.4	25.2	24.7	22.8	20.1	18.5	18.4	18.0	20.3	21.5	23.4	24.6	(29)	
		MCDB	31.7	31.5	29.5	29.5	27.1	25.1	25.4	26.7	29.7	30.5	31.1	31.8	(30)	
	5%	WB	24.9	24.7	24.1	21.8	19.0	17.5	16.7	16.7	19.1	20.7	22.6	24.0	(31)	
		MCDB	31.1	31.3	29.7	29.0	26.7	24.1	23.8	25.7	29.1	30.7	30.5	31.5	(32)	
	10%	WB	24.3	24.2	23.3	20.7	18.0	16.5	15.4	15.6	17.9	19.8	21.9	23.4	(33)	
		MCDB	30.8	31.1	29.9	28.3	25.8	22.9	23.3	25.2	28.7	30.9	30.3	31.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.1	12.2	12.6	13.4	13.7	13.5	14.7	15.9	16.1	15.2	14.2	13.1	(35)	
		MCDBR	14.2	14.0	14.3	14.7	14.6	14.7	16.3	17.9	17.8	17.0	15.9	15.3	(36)	
	5% WB	MCWBR	4.7	4.3	4.9	6.1	6.8	7.2	8.3	8.8	8.0	7.1	6.3	5.8	(37)	
		MCWBR	10.4	10.9	10.4	11.8	12.4	12.6	13.2	14.9	14.2	13.8	12.5	11.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.375	0.357	0.345	0.331	0.310	0.307	0.319	0.362	0.378	0.387	0.399	(40)	
		taud	2.466	2.495	2.526	2.509	2.489	2.539	2.525	2.469	2.366	2.346	2.369	2.378	(41)	
	Ebn at Noon		959	955	947	917	890	892	908	930	927	941	951	947	(42)	
		Edh at Noon	120	114	106	100	93	85	89	102	122	131	131	131	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.16	7.08	6.50	5.69	4.68	4.17	4.61	5.57	6.52	7.33	7.53	7.41	(44)	
	RadStd		0.59	0.40	0.48	0.47	0.24	0.29	0.24	0.23	0.47	0.45	0.40	0.47	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page